Fish Video Analysis Procedure

This procedure outlines how to analyze stationary unbaited remote underwater videos (RUVs) to quantify species richness and relative abundance using the metric MeanCount for fish community assessments. These metrics provide a standardized estimate of relative fish abundance and will be used to compare long-term monitoring of the living shoreline offsetting project.

To reduce processing time, videos are subsampled to assess fish communities. MeanCount is derived by systematically selecting individual frames from across the video to calculate the mean number of fish observed per frame.

Commented [OB1]: Laboratory studies have shown that MeanCount is linearly related to true abundance (Bacheler, N., Shertzer, K, 2015; Erickson, K., Bugnot, A., Figueira, W., 2023) ; however field and simulation studies have shown that it can be less precise and potentially over-inflate zero counts (Erickson, K., Bugnot, A., Figueira, W., 2023)

Video Analysis Process

1. **Accessing Videos**

Videos are stored on the external hard drive listed as “My Passport (E:)” when connected to a computer or laptop.

- Navigate to the “Fish and Fish Habitat Monitoring folder”. Select the year, season and date of the monitoring.
- Videos are organized by transect and STAT number
- Open the video in VLC Media Player or another media player that provides high quality resolution, and allows you to mute the video sound and adjust the speed as needed

*Note: Some videos may have been automatically split into multiple files. Ensure the video filename matches the Video ID listed in the Excel spreadsheet before beginning analysis.

2. Selecting Video Segment for analysis

Of each ~ 20-minute video, a 15-minute segment will be analyzed.

- Discard the first and last minute of the video to allow sediment disturbance to settle
- If sediment disturbance persists or the camera is still being set up, up to 4 minutes before beginning the analysis window can be removed

Record the start and end times used for the analysis

Visibility Assessment

Before beginning analysis of each video for the 2025 surveys, review the associated visibility video and estimate horizontal visibility distance.

Have 2 – 3 independent observers assign one of the following visibility categories:

- <0.5 m
- 0.5–1.0 m
- 1.0–1.5 m
- 1.5 m

Some videos have a visibility pole that have been placed at a known distance from the camera. Determine if the visibility is greater or less than the know distance and assign it to a visibility category. Continue to monitor the visibility at each subsample. Note if the visibility is increasing or decreasing

For future videos

A 2m PVC pole with an attached measuring tape and black electrical tape wrapped around the pole every 0.5m is connected to the bottom of the camera frame mount and extends horizontally along the sea floor.

- Using the measuring tape and barring on the pole, record the distance you can still clearly make out the numbers on the measuring tape, and assign one of the following visibility categories.
    - <0.5 m
    - 0.5–1.0 m
    - 1.0–1.5 m
    - 1.5 m

Video Review and MeanCount Analysis

Watch the selected analysis window in real time and record all species observed throughout the video. Species should be identified to the lowest practicable taxonomic level.

Relative abundance will be estimated using MeanCount derived from systematic subsampling, meaning frames are selected at equal intervals throughout the analysis window. The mean number of fish observed across the frames = MeanCount

For each video:

- Analyze a 15-minute segment
- Sub-samples will be taken systematically every 60 seconds for each video, for a total of 15 frames.

- One second = one analyzed frame
- Watching the frame section multiple times and at slower playback speeds (e.g., 0.25× speed) is encouraged to help ensure fish are not missed.

Counting Guidelines

Count how many fish and invertebrates (crabs, shrimp, worms, snails etc) are observed in the sub-sample frame.

To help interpret and provide consistency in counting and identifying fish within each sub-sampled frame:

- Watch approximately 5 seconds before and 5 seconds after the selected timestamp to provide some context to the frame.
    - Fish are not always easily observed when viewing only the exact one-second frame. Some fish may already be present in the frame but difficult to detect due to being camouflaged on the ocean bottom, poor visibility resulting from turbidity, sediment, algae, lighting etc.,

Commented [OB2]: Potentially increase this to 10seconds. The suggestion is because species like flounder are easy to miss unless you see them swimming or moving along the bottom.

- Only fish observed within the designated one-second frame should be included in the total count.
    - If a fish becomes visible within approximately 0.5 seconds and no more than 1 second on either side of the sub-set point when slowed down to 0.25x speed, it may be included in the count.
    - The idea to take the count from only one frame is to avoid double-counting fish, so if you can say with 100% certainty that is was a different fish than what was counted in the frame, it can be included.

Example:

- If the sub-set point occurs at 3:40 and a fish becomes visible at 3:39/3:40 or 3:40/3:41 when viewed at 0.25× speed, it may be included in the count.
- If the fish entering or exiting the frame is not visible until the full second before or after the subsample point ie 3:39 or 3:41, it should not be included in the 3:40 count, although a note may be added.

Fish can be very difficult to detect when water is turbid (e.g., suspended algae or sediment reflecting light) or when footage is dark, blurred, or cloudy.

Fish and invertebrates should only be included in the count if they are visible and can be distinguished as an organism based on at least two of the following characteristics:

- Fish shape
- Anatomical features (e.g., fins, scales, lateral line, eyes, etc.)
- Colouration or flashes/reflections of scales
- Swimming behaviour or movement patterns

In some cases, a school of fish pass through the frame, but only a few of the individuals can be clearly observed or identified based on the characteristics above, while the remaining movement may appear only as motion in the background. In these situations, only fish that can be identified confidently based on the above characteristics should be included in the count. This may underestimate the true number of fish present within the frame; however, organisms should not be included if there is uncertainty whether the movement represents fish, algae, sediment, bubbles, or other background interference. This is done to reduce the risk of overestimating counts.

- Once confident in the final count, record the number of fish observed in the "Total Fish Count" column for the corresponding sub-set point/row. The same procedure can be followed for recording invertebrate counts.

- If individual species and the number of each are known, add the number of each species observed in the frame to the corresponding species column. If the species is unknown, add the number of each species that are unknown to the "Unknown/Unable to ID due to visibility" column.
- If species with identified, record the total number of species observed in the "Total species observed column

- Add any comments you may have in the "Notes" column. Comments could include anything about visibility, how you determined you count ie characteristics used to determine it was a fish, if you feel there count have been more fish in the frame but they could not be confirmed based on features, potential species present etc

At each subsample frame, record:

- Changes in visibility 0.5m or greater
- Total fish count
- Invertebrate count (if large enough to observe)
- Number of Species present
- Number of individuals per species (if possible)

- Notes or observations etc.

Optional Observations

While reviewing the video, also note whether juvenile fish are present:

- 1 = juveniles present
- 0 = juveniles absent

When done with the Backup drive, select 'Eject' to properly disconnect it before removing the USB.

Video Validation – Quality Control

- All videos will undergo a validation review
- A Random review of a minimum of 10% of annotated video frames (6 – 12 frames per video) from each video
- If agreement is below 95% for all count estimates, reannotation should be undertaken.
- Randomly selected frames should be blind counted at least twice, meaning that two people count the same subsample without knowing the other person's count. Each person's count should be recorded and the average taken to determine the estimated number of fish in each frame. There should be less than 6% difference between the two counts
    - To find the percent difference
    - If the counts have a difference of higher then 6%, the frame will need to be recounted at least once more, preferably by a different counter (keep track of who does each count)

Commented [OB3]: small dataset

Commented [OB4]: 15 - 20% because of the small dataset?

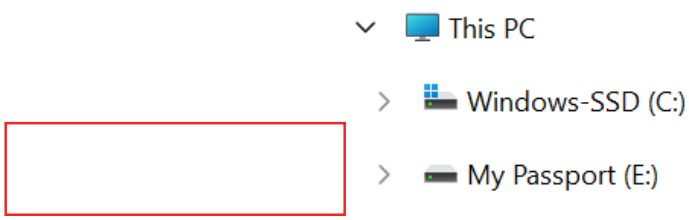
Commented [OB5]: for all videos?

Commented [OB6]: from (Langlois et al., 2020)

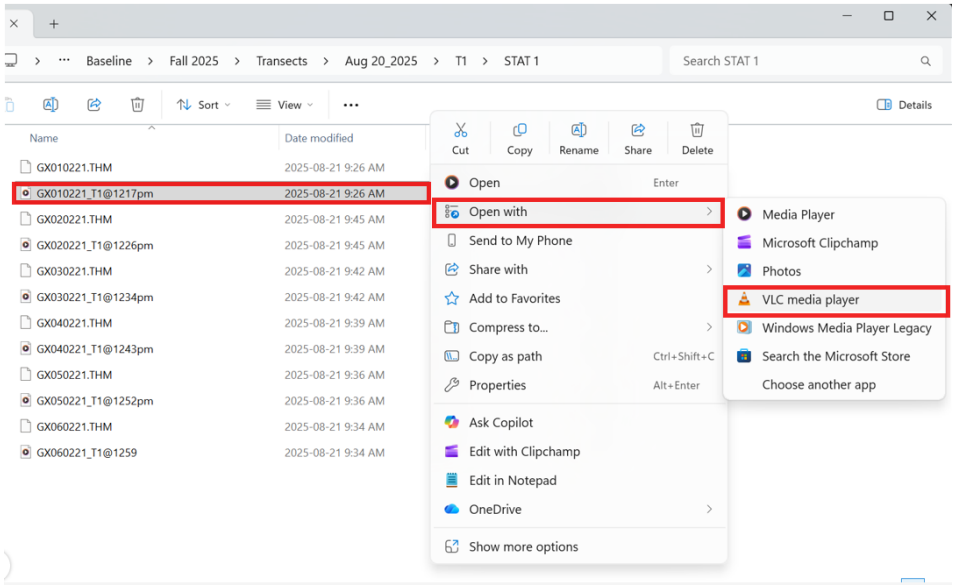
Commented [OB7]: from CA's elver protocol

How to:

- Videos are stored on the hard drive listed at "My Passport (E:) when connected to a computer or laptop.



- Click on the “**Fish and Fish Habitat Monitoring 2025**” folder -> “**Baseline**” folder -> “**Fall 2025**” folder -> “**Transects**” folder
- Within the “**Transects**” folder, the videos have been separated by the date recorded
    - All STAT 1 videos can be found in the “**August 20_2025**” folder, and all STAT 2 & STAT 3 videos can be found in the “**August 27_2025**” folder
    - Within these folders, the videos can be found in the corresponding transect folders ie T1, T2, T3 etc. and STAT #
    - Some videos have been automatically split into smaller videos so make sure to double check the video name match on the hard drive matches the video ID in Excel document
- Open the Excel folder named “”
    - The Excel identifies the transect and STAT #, video ID, sub- sample number and video time that need to be validated
    - 6 sub-points have been randomly selected to be validated from each video
- Starting with the first sub-set point (eg,) navigate and open the corresponding video
    - Open the video with the VLC media player –
    - Once opened, go to the corresponding time for the subset



- To validate a point, count how many fish and invertebrates (crabs, shrimp, worms, snails etc) that are observed in that 1 second period of time
    - Because some fish may have entered the frame previously but are camouflaged on the ocean bottom or the video visibility is poor, they may not be easily observed when just looking at that one second period. To help give context to the frame, watch 5 seconds before the sub-set point and 5 seconds after the sub-set point but only count the observed fish on the sub-set point. If a fish was observed within 0.5 – 1 second on either side of the sub-set point, it can be included. Ie. the sub-set point is at 3:40 and a fish is observed at 3:39/3:40 or 3:40/3:41 when slowed down to 0.25 speed then it can be included in the count. If the fish is not observed until fully at 3:41 then do not include it in the count, but make a note of it. The idea is that we do not want to double count fish, so if you can say that with 100% certainty that it was a different fish than what was counted in the frame, it can be included.
    - Fish can be very challenging to spot when the water is turbid (ie lots of algae or sediment in the water column and reflecting light), and the video clarity is poor (dark, blurred or cloudy)
    - Fish and invertebrates are only to be included if they are visible and can be determined to be a fish and distinguish based on at least 2 of the following characteristics:
        - Fish shape
        - Anatomical features (fins, scales, lateral line, eye etc)
        - Coloration (this could be a flash of scales)

        - swimming behaviours
    - In some cases, a school of fish may be passing through the frame, but you can only see a few clearly or ID based on characteristics, and the rest just looks like movement in the background. Here, unless you can with 100% confidence say that it is a fish based on the above features, one would only include what they can say with certainty is a fish. In these cases, we are likely getting an underestimate of the number of fish in a frame; however, we cannot rule out the possibility that it's simply algae, sediment or bubbles in the water etc. and we do not want to overestimate our counts either
    - Watching the video multiple times and at a slower speed ie 0.25 speed is encouraged to ensure no fish were missed.
- Once confident in the number of fish observed in the frame, add it to the "Total Fish Count" column of the corresponding subset point/row. The same can be done with the invertebrate count.
- Add any comments you may have in the "Notes" column. Comments could include anything about visibility, how you determined you count ie characteristics used to determine it was a fish, if you feel there count have been more fish in the frame but they could not be confirmed based on features, potential species present etc
    - Example: You need to validate T1-STAT 1, subset point 34 which was take at 3:40. Slide the video time to 3:35. Watch the video leading up to 3:40. At 3:40 you can start your count, but your count would end at the start of 3:41.
- When done with the Backup drive, select 'Eject' to properly disconnect it before removing the USB.

PREVIOUS DRAFT LEFT FOR NOW BECAUSE ASSOCIATED COMMENTS

1.2 **MeanCount**

MeanCount is derived from systematic subsampling – ie points were distributed at equal intervals throughout the video.

Based on recommendations from Erickson et al. (2023) and a study by Baker et al. (2022), sub-samples will be taken systematically every 60 seconds for each video, for a total of 12 frames. The average of the samples

Collect 60 evenly spaced time points per video. 1 second = 1 frame

- 12-min analyzed segment → record every 12 seconds

Commented [OB8]: they will only be 1 MaxN per video

Commented [OB9]: "Accuracy and precision tended to increase when using the systematic subsampling method (over the random) and when removing schooling species." (Erickson et al. (2023) "The species accumulation curve began to plateau at 15 min, at the point of 75% species detection, matching the findings of a previous study on RUVs" (Erickson et al. (2023) " the optimal soak time for a general question regarding abundant or fishery-dependant species, excluding cryptobenthics, would be 15 min. A total of 60 min or more would be appropriate for studies looking at less abundant species." (Erickson et al. (2023) " Systematic sampling has a higher chance of catching these groups of fish that are only in the frame for a few minutes." (Erickson et al. (2023) "e. The 'inflexion point' on the curve for accuracy of all species (Fig. 3A) is at 60 frames, where reducing sampling effort further results in a sharp decrease in accuracy, and an increase in variability (larger standard error bars)." Erickson et al. (2023)

- At each subsampled frame, record:
    - species present (lowest practicable taxonomic level)
    - number of individuals per species (if possible)
    - total fish count

3. Video Processing Workflow

3.1 Visibility Assessment

- Review visibility videos -
- Have 2–3 independent observers estimate horizontal visibility distance.
- Assign a visibility category:
    - <0.5 m
    - 0.5–1.0 m
    - 1.0–1.5 m
    - >1.5 m

3.2 Selecting the Analysis Window

- Analyze 12 minutes of each video.
- Discard the first and last minute to allow sediment to settle.
- If the video is >15 minutes, use only the first 15 minutes.
- If sediment requires more time to settle, allow up to 4 minutes before starting the analysis window.

3.4 MeanCount Subsampling

- While watching the video in real time, note subsample timestamps at evenly spaced intervals (total = 60).

At each interval:

- Pause and record species, counts, and total count

3.6 Juvenile Presence (Optional but Recommended if possible)

While reviewing the video in real time, note whether juveniles are present.

- Assign a binary category per video:
    - 1 = juveniles present
    - 0 = not present

Commented [OB10]: A note on horizontal visibility from a hardcopy draft of DFO's *WCVI Snorkel Survey Protocols for salmon enumeration in West Coast Vancouver Island streams* June19, 2014 : "Secchi readings should not be considered the same as effective visibility distance (EVD), which is typically 50 - 70% of the secchi distance. EVD is the distance at which fish can be identified and species size and sex reliably determined."

Commented [OB11]: if sediment has not settled or the camera is still in the process of being set up up, up to 10 minutes can be removed for videos longer than 25 minutes

Commented [OB12R11]: up to 15 minutes for videos longer than 30minutes

4. Summary Workflow

- Review visibility
- Select the 12 minute analysis window
- record species richness
- capture screenshots
- Conduct subsampling (60 points)
- record MeanCount data
- Note juvenile presence (if possible)

Compile report with all metrics and screenshots

Commented [OB13]: fish are counted on the subset frame but if a fish is observed with 0.5 seconds on with side of the subset point ie subset is at 0:28 and a fish is observed at 0.27/0:28 or 0:28/0:29 when slowed down to 0.25 speed then it can be included in the count. if the fish is not observed until fully at 0:29 then do not include in count but can make a note of it

References (Not formatted)

- Bacheler, N., Shertzer, K. Estimating relative abundance and species richness from video surveys of reef fishes. Fish Bull. 113:15–26 (2015). doi: 10.7755/FB.113.1.2 https://spo.nmfs.noaa.gov/sites/default/files/bacheler.pdf
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- Baker, R., Bilbrey, D., Bland, A. et al. (2022). Underwater video as a tool to quantify fish density in complex coastal habitats. https://www.mdpi.com/1424-2818/14/1/50
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    - https://benthic-bruvs-field-manual.github.io/files/Chapter-05_Benthic-Stereo-BRUVs_Field-Manual_V3.pdf